Year 7

REVISION PACK

Unit 1

Name: _____

Class: _____

Unit 1 · Averages and range

Worked example: mode, median, mean and range

Pets owned by 8 students: 3 1 4 4 2 6 4 0

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

(a)	**Step 1**	Most common number	4 (3 times)
	Step 2	Answer	4
(b)	**Step 1**	Put in order	$0, 1, 2, 3, 4, 4, 4, 6$
	Step 2	Middle two	3 and 4
	Step 3	Add, then divide by 2	$(3 + 4) \div 2 = 3.5$
(c)	**Step 1**	Add all the numbers	$3 + 1 + 4 + 4 + 2 + 6 + 4 + 0 = 24$
	Step 2	Divide by how many (8)	$24 \div 8 = 3$
(d)	**Step 1**	Biggest − smallest	$6 - 0 = 6$

Practice: mode, median, mean and range

1.1 Bags carried by 8 shoppers: 2 0 3 1 3 5 3 7

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

1.2 Pets owned by 8 families: 1 4 2 4 0 4 3 6

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

1.3 Goals scored in 8 football matches: 3 1 6 1 0 1 4 8

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

1.4 Text messages sent by 8 students: 12 7 9 12 5 14 12 9

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

1.5 Cars in 8 car parks: 20 25 18 25 30 22 25 19

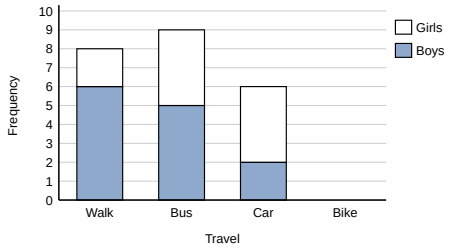
(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

1.6 Children in 8 families: 2 5 3 2 1 4 2 5

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

Unit 1 · Compound bar charts

Worked example: completing and reading a compound bar chart



Some students were asked how they travel to school. 3 boys and 4 girls come by bike.

(a) Complete the compound bar chart.

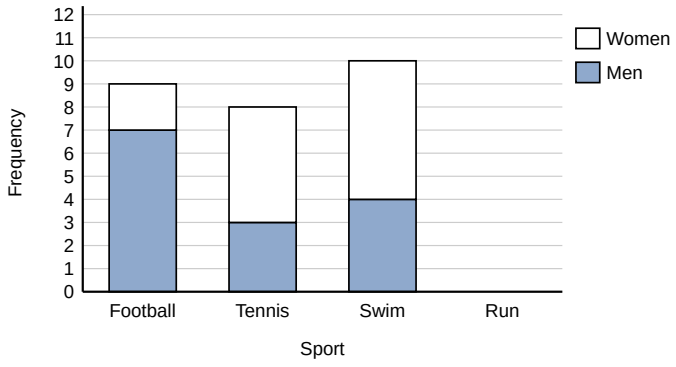
(b) How many girls come by bus?

(c) Work out the total number of boys.

(a)	Step 1	Draw the boys bar	up to 3
	Step 2	Girls on top	$3 + 4 = 7$
	Step 3	Answer	Bike bar: shade to 3, white from 3 to 7
(b)	Step 1	Top of the bar	9
	Step 2	Top of the boys	5
	Step 3	Top of bar − boys	$9 - 5 = 4$
(c)	Step 1	Read each boys bar	$6,\ 5,\ 2,\ 3$
	Step 2	Add the boys bars	$6 + 5 + 2 + 3 = 16$

Practice: completing and reading a compound bar chart

2.1



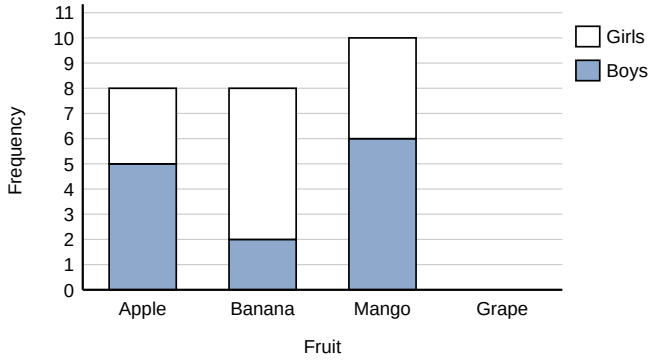
People were asked what sport they play. 5 men and 3 women chose run.

(a) Complete the compound bar chart.

(b) How many women chose swim?

(c) Work out the total number of men.

2.2



Students were asked their favourite fruit. 3 boys and 4 girls chose grape.

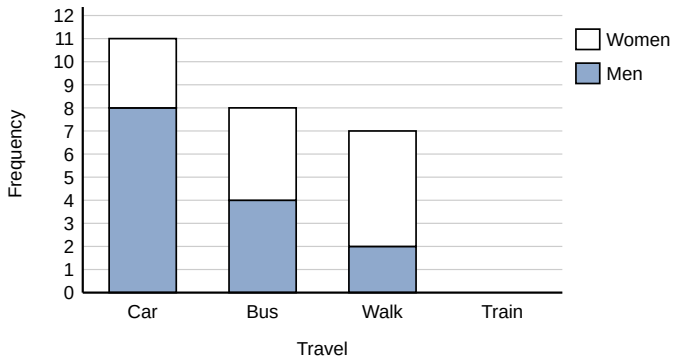
(a) Complete the compound bar chart.

(b) How many girls chose banana?

(c) Work out the total number of boys.

Practice: completing and reading a compound bar chart (continued)

2.3



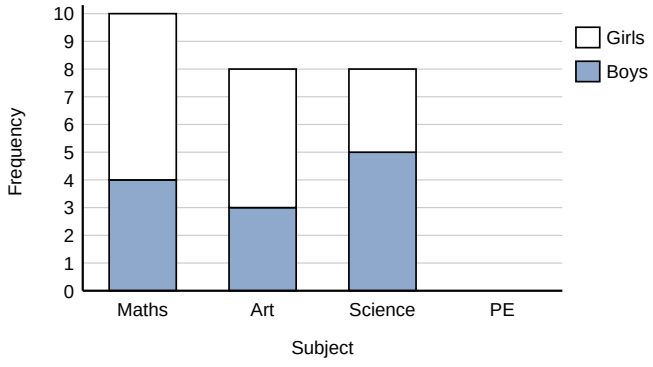
People were asked how they get to work. 3 men and 6 women chose train.

(a) Complete the compound bar chart.

(b) How many women chose car?

(c) Work out the total number of men.

2.4



Students were asked their favourite subject. 6 boys and 2 girls chose pe.

(a) Complete the compound bar chart.

(b) How many girls chose art?

(c) Work out the total number of boys.

Worked example: the mean and adding a new value

Sweets in 5 bags: 8 5 7 4 6

(a) Work out the mean.

(b) A sixth bag has 2 sweets. Will the mean for all 6 bags be greater or less? Give a reason.

(a)	Step 1	Add all the numbers	$8 + 5 + 7 + 4 + 6 = 30$
	Step 2	Divide by how many (5)	$30 \div 5 = 6$
	Step 3	Answer	6
(b)	Step 1	Compare 2 with the mean	$2 < 6$
	Step 2	Answer	Less, because 2 is less than the mean

Practice: the mean and adding a new value

3.1 Children in 5 houses: 3 1 5 2 4

(a) Work out the mean.

(b) A sixth house has 0. Will the mean for all 6 be greater or less? Give a reason.

3.2 Goals in 5 matches: 2 4 1 3 5

(a) Work out the mean.

(b) A sixth match has 6. Will the mean for all 6 be greater or less? Give a reason.

3.3 Books read by 5 students: 6 9 4 7 9

(a) Work out the mean.

(b) A sixth student has 1. Will the mean for all 6 be greater or less? Give a reason.

3.4 Marks of 5 students: 12 15 10 14 9

(a) Work out the mean.

(b) A sixth student has 20. Will the mean for all 6 be greater or less? Give a reason.

3.5 Eggs from 5 hens: 5 7 4 6 8

(a) Work out the mean.

(b) A sixth hen has 2. Will the mean for all 6 be greater or less? Give a reason.

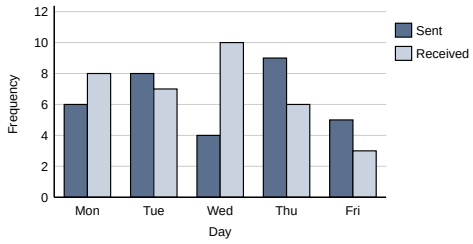
3.6 Minutes late for 5 trains: 4 0 6 3 2

(a) Work out the mean.

(b) A sixth train has 9. Will the mean for all 6 be greater or less? Give a reason.

Unit 1 · Dual bar charts

Worked example: reading and comparing a dual bar chart



The dual bar chart shows the emails Omar sent and received on 5 days.

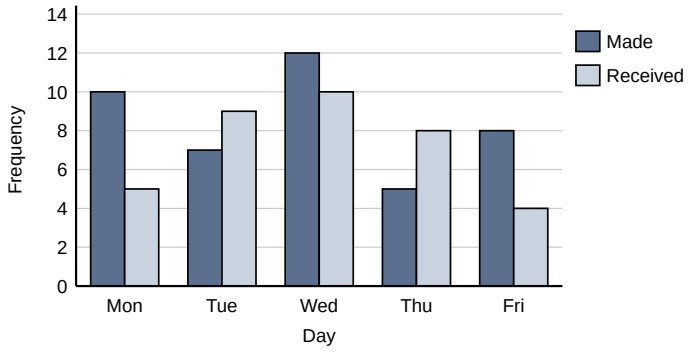
(a) On which day was the difference between sent and received the greatest?

(b) Omar says, "I sent more emails than I received." Is Omar correct? Show your working.

(a)	Step 1	Find each difference	$2,\ 1,\ 6,\ 3,\ 2$
	Step 2	Answer	Wednesday
(b)	Step 1	Add the sent	$6 + 8 + 4 + 9 + 5 = 32$
	Step 2	Add the received	$8 + 7 + 10 + 6 + 3 = 34$
	Step 3	Compare	$32 < 34$
	Step 4	Answer	No, he received more

Practice: reading and comparing a dual bar chart

4.1

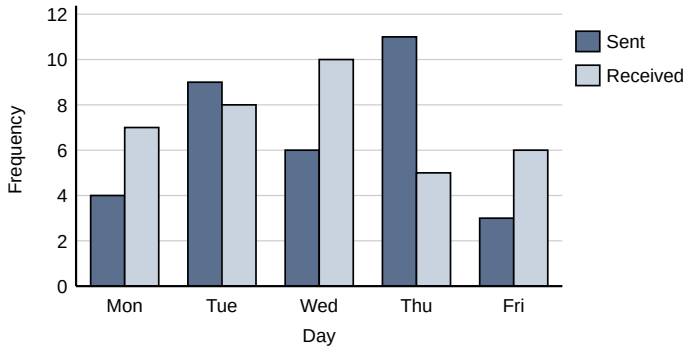


The dual bar chart shows the number of calls Jamila made and received on 5 days.

(a) On which day was the difference between made and received the greatest?

(b) Jamila says, "I made more calls than I received." Is Jamila correct? Show your working.

4.2



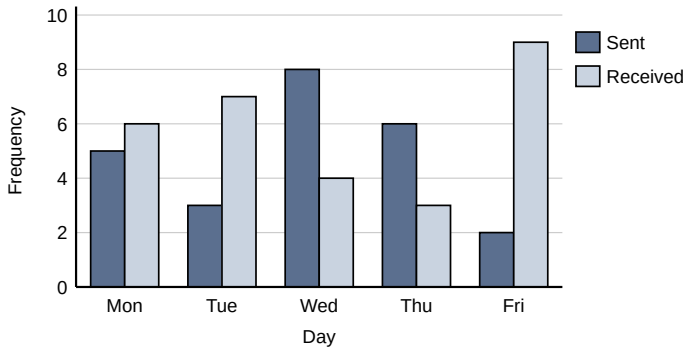
The dual bar chart shows the number of texts Sami sent and received on 5 days.

(a) On which day was the difference between sent and received the greatest?

(b) Sami says, "I sent more texts than I received." Is Sami correct? Show your working.

Practice: reading and comparing a dual bar chart (continued)

4.3

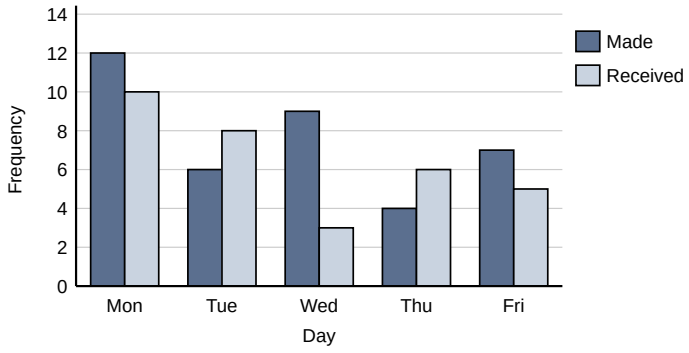


The dual bar chart shows the number of emails Lina sent and received on 5 days.

(a) On which day was the difference between sent and received the greatest?

(b) Lina says, "I sent fewer emails than I received." Is Lina correct? Show your working.

4.4



The dual bar chart shows the number of calls Yusuf made and received on 5 days.

(a) On which day was the difference between made and received the greatest?

(b) Yusuf says, "I made more calls than I received." Is Yusuf correct? Show your working.

Worked example: comparing with the median and range

Sam's points in 7 games: 4 8 5 10 6 7 5

(a) Work out the median and the range.

(b) Ella: median 8, range 3. Compare Sam and Ella.

(a)	Step 1	Put in order	$4, 5, 5, 6, 7, 8, 10$
	Step 2	Median: the middle	6
	Step 3	Range: biggest − smallest	$10 - 4 = 6$
(b)	Step 1	Compare the medians	$8 > 6$
	Step 2	Say what it means	Ella did better on average
	Step 3	Compare the ranges	$3 < 6$
	Step 4	Answer	Ella is more consistent

Practice: comparing with the median and range

5.1 Kareem's points in 7 basketball games: 5 2 9 6 4 8 6

(a) Work out the median and the range.

(b) Aziz: median 8, range 15. Compare Kareem and Aziz.

5.2 Hana's points in 7 netball games: 14 18 11 16 20 13 17

(a) Work out the median and the range.

(b) Rania: median 12, range 4. Compare Hana and Rania.

5.3 Tariq's goals in 7 football matches: 2 6 3 8 5 1 4

(a) Work out the median and the range.

(b) Faris: median 6, range 3. Compare Tariq and Faris.

5.4 Dana's marks in 7 quizzes: 30 25 34 28 40 27 31

(a) Work out the median and the range.

(b) Mona: median 26, range 20. Compare Dana and Mona.

5.5 Ibrahim's points in 7 basketball games: 9 13 10 7 12 15 11

(a) Work out the median and the range.

(b) Khalid: median 9, range 3. Compare Ibrahim and Khalid.

5.6 Nadia's goals in 7 hockey matches: 6 8 4 9 7 5 10

(a) Work out the median and the range.

(b) Salma: median 9, range 12. Compare Nadia and Salma.

Worked example: grouped frequency tables

The heights of 12 plants, in cm: 23 35 18 27 31 14 26 38 22 29 25 33

(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 20 cm?

(a)	Step 1	Tally each number	$10 < h \leq 20:\ 2$
	Step 2		$20 < h \leq 30:\ 6$
	Step 3		$30 < h \leq 40:\ 4$
(b)	Step 1	Bars, no gaps	heights $2,\ 6,\ 4$
(c)	Step 1	Biggest frequency	$20 < h \leq 30$
(d)	Step 1	Add the groups above 20	$6 + 4 = 10$
	Step 2	Answer	10

Practice: grouped frequency tables

6.1 The masses of 15 apples, in g:

112 145 130 168 155 121 139 147 176 163 158 141 126 149 137

M (g)	Tally	Frequency
100 < M ≤ 120		

(a) Complete the frequency table. Use groups of width 20.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 140 g?

6.2 The times taken by 16 runners, in minutes:

23 35 41 28 37 32 45 26 39 31 34 22 38 43 36 29

T (minutes)	Tally	Frequency
20 < T ≤ 30		

(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 30 minutes?

Practice: grouped frequency tables (continued)

6.3 The masses of 18 potatoes, in g:

212 358 305 245 419 337 315 320 284 180 312 390 187 289 336 340 332 310

M (g)	Tally	Frequency
100 < M ≤ 200		

(a) Complete the frequency table. Use groups of width 100.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 300 g?

6.4 The heights of 15 students, in cm:

142 156 149 161 153 138 157 165 151 147 159 154 144 152 158

H (cm)	Tally	Frequency
130 < H ≤ 140		

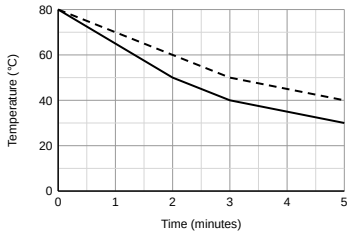
(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 150 cm?

Worked example: reading and comparing line graphs



Two cups of water cool down. The graph shows their temperature every minute. — Cup A - - - Cup B

(a) Write down the temperature at the start.

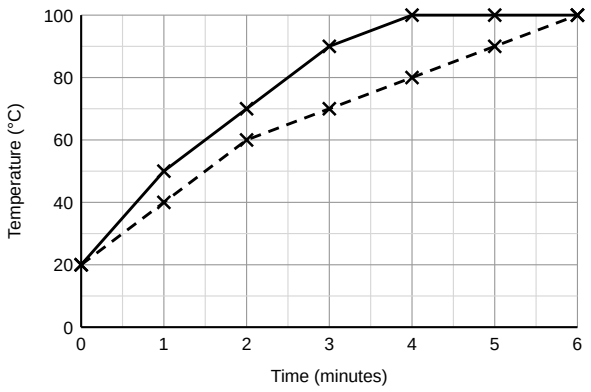
(b) Compare the times taken for each cup to reach 40 °C.

(c) Work out the difference in temperature after 2 minutes.

(a)	Step 1	Read at 0 minutes	$80\ ^\circ\text{C}$
(b)	Step 1	Cup A reaches 40	3 minutes
	Step 2	Cup B reaches 40	5 minutes
	Step 3	Answer	A is 2 minutes quicker
(c)	Step 1	Read both at 2 minutes	50 and 60
	Step 2	Higher − lower	$60 - 50 = 10\ ^\circ\text{C}$

Practice: reading and comparing line graphs

7.1



Two liquids are heated from room temperature. The line graph shows their temperature every minute.

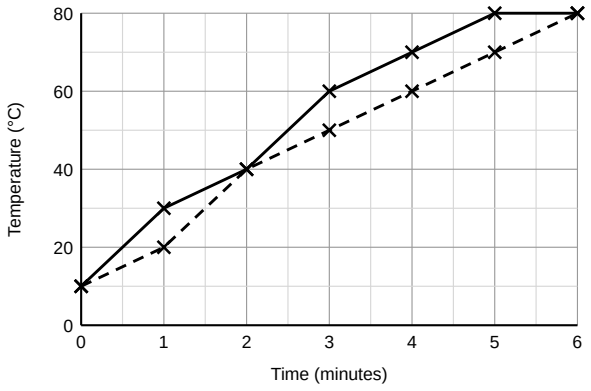
— Liquid A - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 100 °C.

(c) Work out the difference in temperature after 3 minutes.

7.2



Two liquids are heated. The line graph shows their temperature every minute.

— Liquid A - - - Liquid B

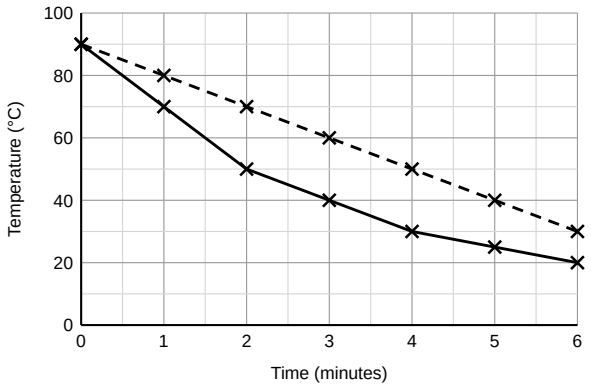
(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 60 °C.

(c) Work out the difference in temperature after 2 minutes.

Practice: reading and comparing line graphs (continued)

7.3



Two liquids are left to cool. The line graph shows their temperature every minute.

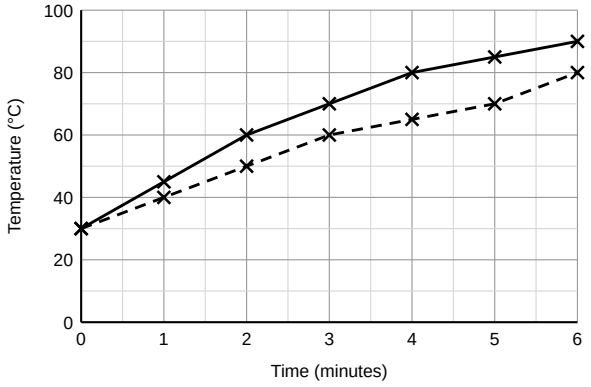
— Liquid A - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 50 °C.

(c) Work out the difference in temperature after 4 minutes.

7.4



Two liquids are heated. The line graph shows their temperature every minute.

— Liquid A - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 60 °C.

(c) Work out the difference in temperature after 4 minutes.